

BYDGOSZCZ, Poland

WMO: 122400

Lat: 53.0964N

Lon: 17.9950E

Elev: 72

StdP: 100.46

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.2	-18.1	0.8	-14.9	-15.1	1.0	-11.1	12.2	6.1	11.0	4.6	1.9	90	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.8	19.6	28.9	18.9	27.0	18.2	21.2	27.7	20.4	26.3	19.4	24.8	4.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.1	22.9	18.2	13.2	22.4	17.2	12.4	21.7	61.9	27.8	58.8	26.2	55.4	24.9	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.6	8.4	7.4	-19.1	33.7	5.4	2.0	-23.0	35.1	-26.2	36.2	-29.2	37.3	-33.2	38.7		
			-19.4	22.8	5.4	0.8	-23.2	23.3	-26.4	23.8	-29.4	24.2	-33.2	24.8		
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-1.8	-0.7	3.2	8.9	13.4	17.0	19.5	18.9	14.3	8.8	4.7	1.0
	DBStd	8.42	5.70	4.92	4.65	4.09	4.14	3.46	3.25	3.12	3.37	3.82	3.51	4.74
	HDD10.0	1471	364	299	213	66	13	1	0	0	4	69	161	281
	HDD18.3	3557	623	532	468	282	160	66	25	31	127	296	409	539
	CDD10.0	1106	0	0	3	35	121	209	296	276	134	31	2	0
	CDD18.3	151	0	0	0	1	8	24	62	48	7	0	0	0
	CDH23.3	1606	0	0	0	22	121	308	596	474	86	0	0	0
	CDH26.7	463	0	0	0	6	23	86	190	137	22	0	0	0
Wind	WSAvg	3.4	3.8	3.4	3.7	3.5	3.3	3.4	3.2	2.9	3.0	3.3	3.4	3.8
Precipitation	PrecAvg	541	34	26	36	30	54	59	79	63	48	37	36	38
	PrecMax	692	84	60	72	87	98	112	138	161	119	122	104	78
	PrecMin	384	4	5	9	2	16	20	8	14	12	5	4	7
	PrecStd	87	18	12	17	17	21	27	36	34	26	28	17	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	11.2	17.9	26.1	29.1	32.1	33.2	33.0	29.2	21.8	14.1	11.2
		MCWB	8.3	8.2	11.2	14.4	17.8	20.9	20.7	20.9	18.6	15.1	12.3	9.6
	2%	DB	8.2	9.1	14.8	22.1	26.8	29.0	30.9	30.0	25.8	19.0	12.2	9.8
		MCWB	6.9	6.9	9.0	13.4	17.1	18.7	19.7	19.8	17.0	14.2	10.3	8.8
	5%	DB	6.8	7.1	12.2	19.2	24.1	26.9	28.9	27.9	23.1	16.8	11.0	8.1
		MCWB	5.7	5.2	8.1	11.5	16.0	17.5	19.2	18.8	16.1	13.4	9.4	7.0
	10%	DB	5.2	5.2	10.1	17.0	21.9	24.8	26.9	26.0	21.1	14.9	9.8	6.8
		MCWB	4.3	3.9	7.0	10.7	14.8	16.7	18.7	18.0	15.3	12.3	8.8	5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	9.1	12.2	15.9	19.2	22.1	22.3	22.1	19.8	16.4	13.2	10.2
		MCDB	9.9	10.2	16.6	23.4	26.1	28.9	28.6	28.7	27.9	20.1	14.2	11.1
	2%	WB	6.9	7.0	10.2	14.1	17.8	20.3	21.2	20.9	18.2	14.9	10.8	8.6
		MCDB	7.9	8.4	13.1	20.5	24.4	26.4	27.5	27.5	24.0	17.9	12.0	9.4
	5%	WB	5.6	5.3	8.7	12.6	16.7	18.8	20.6	20.0	17.0	13.6	9.7	6.9
		MCDB	6.5	6.7	11.5	18.0	22.8	24.8	26.4	25.6	21.6	16.2	10.6	7.8
	10%	WB	4.3	4.0	7.3	11.2	15.5	17.6	19.6	19.0	15.8	12.5	8.6	5.7
		MCDB	5.1	5.1	9.7	16.0	20.8	22.8	24.8	23.8	20.0	14.7	9.5	6.4
Mean Daily Temperature Range	5% DB	MDBR	4.8	6.3	8.7	11.6	11.4	11.5	10.9	11.3	10.6	8.3	5.2	4.7
		MCDBR	4.7	8.0	12.2	15.4	14.6	14.9	14.1	14.4	14.1	11.6	6.7	5.5
		MCWBR	4.2	6.1	7.7	8.2	6.8	6.5	5.3	5.9	6.8	7.3	5.3	4.7
	5% WB	MCDBR	4.7	6.9	10.7	13.7	12.8	12.5	11.4	11.8	12.3	9.7	6.2	5.3
		MCWBR	4.2	5.6	7.2	7.8	6.6	6.4	5.3	5.5	6.6	6.6	5.4	4.7
Clear-Sky Solar Irradiance	taub	0.311	0.340	0.373	0.418	0.413	0.406	0.437	0.428	0.385	0.365	0.346	0.308	
	taud	2.266	2.271	2.242	2.175	2.242	2.290	2.220	2.248	2.346	2.385	2.332	2.292	
	Ebn at Noon	653	751	802	810	837	846	808	792	785	718	608	584	
	Edh at Noon	63	86	108	131	129	124	131	121	97	77	61	52	
All-Sky Solar Radiation	RadAvg	0.62	1.31	2.53	4.15	5.18	5.41	4.99	4.37	3.16	1.71	0.69	0.44	
	RadStd	0.10	0.20	0.35	0.54	0.55	0.48	0.61	0.41	0.40	0.25	0.09	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (35 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	-155	+71	N/A

Nomenclature: See separate page